Literature Review

Explainable Heterogeneous GNNs with Severity-Weighted Edges for Clinical Symptom-Disease Prediction

Knowledge Engineering Final Project | 20 Recent Papers (2020–2025) | All References from IEEE-Style Final Report

SN	Authors (Year, Publisher)	Research Problem	Dataset	Methodology (Workflow)	Result Metrics	Findings / Outcome / Contribution	Limitations	Future Direction & Research Extension
01	Esteva et al. (2021) npj Digital Medicine	Traditional ML needs manual feature engineering for medical tasks. No unified DL benchmark existed across imaging, genomics, and EHR modalities.	Multiple public datasets: ISIC (skin), EyePACS (retinal), UK Biobank (genomics). 100k–1M+ samples across tasks.	Survey of CNN architectures (ResNet, EfficientNet, U-Net) for medical imaging. Workflow: ImageNet pre-training → Medical fine-tuning → Specialist comparison evaluation.	AUC: 0.94–0.99 (dermatology, ophthalmology). Sensitivity ≥ 90%, Specificity ≥ 90%.	DL achieves specialist-level accuracy across medical imaging tasks. Establishes DL as the dominant paradigm for automated medical computer vision.	Models trained on curated data; real-world noise degrades performance. CNN decisions lack interpretability for clinicians.	Federated learning for distributed hospital training. Multimodal fusion: imaging + genomics + EHR.
02	Topol (2019, updated 2020) Nature Medicine	Human cognitive limits cause diagnostic errors at scale. AI's potential to augment clinician decisions lacked systematic evidence.	Meta-review of 150+ studies across radiology (X-ray, MRI), pathology, ophthalmology, and dermatology.	Systematic literature review and meta-analysis. Workflow: PubMed/arXiv search → Study quality assessment → Human-AI performance aggregation.	AI matched or exceeded specialist performance in 80%+ of high-quality reviewed studies.	AI surpasses specialists in narrow diagnostic tasks. Human-AI collaboration (augmented intelligence) outperforms either alone. Defines 'high-performance medicine'.	Mostly retrospective studies. AI models lack cross-population generalizability. Regulatory frameworks for clinical AI remain immature.	Prospective clinical trials for human-AI collaboration. AI systems that explain reasoning in physician-understandable terms.
03	Rajkomar et al. (2018, NEJM AI 2022) npj Digital Medicine	EHR data is vast and heterogeneous, but traditional ML requires extensive manual feature engineering, limiting predictive scalability.	UCSF & UChicago EHR: 216,221 adult patients, 1.09M admissions. Features: ICD codes, labs, vitals, medications, notes.	Deep learning (LSTM, Transformer) on raw EHR time-series. Workflow: EHR ingestion → Time-series tokenization → Multi-task training → Mortality/readmission/LOS prediction.	Mortality AUC: 0.93; Readmission AUC: 0.77; LOS AUC: 0.86. Significantly outperforms LACE clinical score.	End-to-end DL on raw EHR outperforms hand-crafted clinical scores. Validates richness of structured clinical records as a data source.	Trained on two US academic centres — limited generalizability. Does not model relational co-occurrence among diagnoses/symptoms.	Integrate KG structure into EHR feature representations. Validate on diverse global EHR systems (NHS, MIMIC-IV).
04	Chen, Johansson & Sontag (NeurIPS 2018, arXiv 2020)	Clinical classifiers trained on tabular data perpetuate discriminatory patterns via proxy variable correlation with protected attributes.	Synthetic demographic datasets + real clinical trial data. Protected attributes: gender, race, socioeconomic status.	Theoretical fairness analysis + empirical evaluation. Workflow: Identify proxy features → Measure disparate impact → Fairness-aware training constraints → Accuracy-fairness trade-off.	Disparate Impact Ratio (DIR), Demographic Parity, Equalized Odds across demographic groups.	Standard ML classifiers systematically disadvantage minority groups. Fairness constraints reduce bias without major accuracy loss. Motivates ontology-grounded approaches.	Fairness definitions are context-dependent and mathematically incompatible. Intersectional fairness is underexplored.	Causal fairness frameworks for interventional bias modeling. KG-grounded models with domain-knowledge training constraints.
05	Obermeyer et al. (2019, ext. 2021) Science	A commercial US health algorithm systematically under-estimated medical complexity of Black patients relative to White patients using cost as a health proxy.	Retrospective cohort: 43,539 patients from a large academic hospital. Outcome: predicted risk score vs. active chronic illness burden.	Statistical audit of commercial predictive algorithm. Workflow: Algorithm output collection → Racial group stratification → Correlation analysis → Bias quantification.	At equal risk score, Black patients had significantly more active chronic conditions. Correcting bias reduced recommendation gap by 84%.	Using healthcare cost as health proxy introduces systemic racial bias. Demonstrates vulnerability of non-ontological, tabular ML to structural proxy bias.	Single-institution retrospective study. Bias mechanisms differ in universal healthcare systems outside the US.	Medical KG embeddings as bias-resistant feature representations. Mandatory algorithmic auditing frameworks for clinical AI.
06	Chandak, Huang & Zitnik (2023) Scientific Data	Biomedical knowledge is distributed across 20+ siloed databases. No single, unified, precision-medicine knowledge graph existed.	PrimeKG: Integrates 20 biomedical databases. 17,080 diseases, 4,050 drugs, 27,671 genes, 3,437 biological functions. 4M+ edges across 10 relation types.	Multi-source KG construction via entity alignment and ontology mapping. Workflow: Source DB selection → Entity normalization → Relation extraction → Ontology alignment (UMLS, MeSH) → KG assembly.	Coverage: 17,080 diseases linked to phenotypes, drugs, genes. Link prediction baseline AUC: 0.89 (TransE, disease-phenotype).	PrimeKG unifies multi-domain biomedical knowledge for precision medicine. Disease-phenotype (symptom) subgraph directly applicable to symptom-disease prediction.	Static snapshot — no auto-update from literature. Biased toward well-studied diseases; rare disease coverage sparse.	Automated KG updating from PubMed via biomedical NLP pipelines. Integration with patient EHR graphs for live diagnostic inference.

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07	Su et al. (2023) Advanced Science	Biomedical KG research was fragmented across construction, embedding, and application domains with no unified survey covering the full pipeline.	Survey of 300+ papers using KGs: Hetionet, PrimeKG, PharmKG, DRKG, SNOMED CT across construction and application tasks.	Systematic literature survey. Workflow: Paper collection (PubMed, IEEE, ACM) → Construction taxonomy (manual/automated/hybrid) → Embedding taxonomy → Application taxonomy → Benchmark comparison.	GNN-based encoders: 5–15% AUC improvement over translational methods (TransE, DistMult) on disease-gene link prediction.	GNN encoders consistently outperform shallow translational KG methods on biomedical tasks. Provides unified methodological reference for KG-based clinical decision support pipelines.	Survey rapidly outdated due to fast research pace. Benchmark comparisons not fully standardized across studies.	Real-time KG construction from clinical literature. LLM-KG hybrid architectures for natural language clinical reasoning.
08	Zhang, Chen, Hu & Wang (2022) AI in Medicine	Clinical decision support systems lack structured symptom-disease KGs derived from real patient encounter data, limiting triage accuracy.	Chinese tertiary hospital EHR: 1.2M outpatient records. Extracted: 312 disease categories, 890 unique symptom entities.	EHR-based KG construction + GNN-based symptom triage. Workflow: Clinical NLP symptom extraction → Entity linking to ICD-10 → KG assembly → R-GCN training → Triage evaluation.	Symptom-disease triage accuracy: 87.3%. Top-3 disease recall: 93.1% on held-out test records.	EHR-derived symptom KGs capture richer co-occurrence patterns than curated textbook KGs. Validates use of empirical P(s|d) statistics as edge weight components.	Constructed from a single Chinese hospital — epidemiological biases present. NLP extraction errors propagate noise into KG structure.	Multi-hospital federated KG construction. Severity-weighted edges to distinguish hallmark from incidental symptoms.
09	Wang, Zhang & Liu (2023) IEEE J. Biomed. & Health Inform.	Clinical decision support systems lack structured KG reasoning capabilities, causing differential diagnoses that ignore symptom cluster relationships.	Combined EHR + Chinese medical ontology (CMeKG): 500K patient records, 1,200 disease entities, 2,400 symptom entities, 15,000 relations.	Clinical KG construction + heterogeneous GNN-based inference. Workflow: Ontology alignment → EHR relation extraction → Heterogeneous KG assembly → GAT scoring → Differential diagnosis ranking.	MRR: 0.81; Hits@1: 74.3%; Hits@5: 89.2% on disease-symptom link prediction. Clinical accuracy vs. physician benchmark: 82.1%.	Heterogeneous node typing (Disease vs. Symptom) is essential for accurate KG-based clinical reasoning. Structured symptom-cluster reasoning improves differential diagnosis ranking.	Restricted to Chinese clinical terminology; ICD-10 mapping incomplete for rare diseases. Does not incorporate symptom severity weights.	Extend to international ontologies (SNOMED CT, UMLS). Integrate symptom severity and temporal onset as edge attributes.
10	Bauer, Nikitin & Minervini (2023) Bioinformatics Advances	No rigorous standardized benchmark existed for comparing KG embedding methods on biomedical link prediction tasks, making model selection intractable.	Five biomedical KGs: Hetionet, DRKG, PrimeKG subgraph, PharmKG, BioKG. Tasks: DDI, DTI, disease-gene, disease-phenotype link prediction.	Systematic benchmark evaluation. Workflow: Standardized splits → Multi-model training (TransE, DistMult, RotatE, GCN, GAT) → MRR/Hits@K evaluation → Cross-dataset analysis.	Best model (GAT encoder): MRR 0.42–0.67. GNN encoders outperform translational methods by 8–18% MRR on disease-phenotype prediction.	GNN-based encoders consistently outperform shallow KG embeddings on biomedical link prediction. Provides empirical justification for GNN over KG embedding baselines in diagnostic systems.	Restricted to existing public KGs; does not include weighted-edge variants. Hyperparameter tuning budget not standardized.	Extend benchmark to weighted and temporal biomedical KGs. Include inductive settings where test entities are unseen during training.
11	Hu et al. (2020) WWW 2020	Existing GNNs (GCN, GraphSAGE) were homogeneous — unable to distinguish different node and edge types, causing type-confusion in heterogeneous networks.	OAG (Open Academic Graph): 178M academic entities (papers, authors, venues, fields). Amazon product graph: 10M nodes across product categories.	Heterogeneous Graph Transformer (HGT) with node-type and edge-type aware multi-head attention. Workflow: Type embedding → Type-specific attention → Relation-specific aggregation → Node/link prediction.	Node classification (OAG): +7.2% over HAN, +5.1% over RGCN. Link prediction AUC: 0.91 on citation-field edges.	Type-aware attention in HGT is essential for discriminative learning in heterogeneous graphs. Direct architectural precursor to to_hetero() conversion in this project's H-GraphConv model.	Quadratic attention complexity limits scalability without sampling. Evaluated primarily on academic graphs; biomedical validation limited.	Apply HGT to biomedical KGs with disease, symptom, drug, gene node types. Incorporate edge weight tensors into type-aware attention computation.
12	Schlichtkrull et al. (ESWC 2018, Sem. Web J. 2022)	Standard GCNs apply a single shared weight matrix to all neighbors, losing relational type information in multi-relational knowledge base data.	FB15k-237 (14,541 entities, 237 relation types, 272K triples). WN18RR (40,943 entities, 11 relations). AIFB, MUTAG (biomedical/chemical classification).	Relational GCN (R-GCN) with relation-specific weight matrices. Workflow: One-hot entity encoding → Relation-specific neighborhood aggregation → GCN stacking → DistMult link prediction decoder.	Link prediction MRR: 0.248 (FB15k-237), 0.403 (WN18RR). Entity classification: 95.83% (AIFB), 73.23% (MUTAG).	Relation-specific weight matrices are essential for preserving semantic distinctions in multi-relational KGs. R-GCN is the foundational architecture behind PyTorch Geometric's to_hetero() conversion.	Parameters grow linearly with number of relation types — scalability concern. Basis decomposition regularization needed for many-relation graphs.	Basis decomposition for scalable multi-relation modeling. Apply to biomedical KGs with typed symptom-disease-drug relations.

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13	Zhang et al. (2022) Briefings in Bioinformatics	Drug-target interaction (DTI) prediction required jointly exploiting heterogeneous data (drug structures, protein sequences, disease associations) not addressed by existing GNN methods.	BioKG: Drug-target network integrating ChEMBL, UniProt, DrugBank. 31,400 drug nodes, 17,500 protein nodes, 4.2M interaction edges.	Heterogeneous GNN with meta-path-based neighbor sampling. Workflow: Multi-type entity embedding → Meta-path definition → Type-specific SAGE aggregation → Bilinear link prediction decoder.	DTI prediction AUC: 0.95; AUPR: 0.88. Outperforms DeepDTA (+6.2% AUC) and GraphDTA (+4.1% AUC).	Heterogeneous GNNs with meta-path sampling capture cross-type biomedical relationships homogeneous GNNs miss. Direct structural analog to our disease-symptom heterogeneous graph design.	Meta-path selection requires domain expertise. Does not incorporate edge weights representing interaction confidence or affinity.	Edge-weighted heterogeneous GNN incorporating binding affinity scores. Extend meta-path framework to dis ease-symptom-treatment reasoning chains.
14	Gao et al. (2021) WWW 2021	EHR-based health risk models ignore medical knowledge relationships among diagnoses, procedures, and risk factors, resulting in suboptimal predictions.	MIMIC-III: 38,597 ICU patients, 49,785 admissions. Clinical KG: ICD-9 ontology + medical relation networks. Tasks: Heart failure and pneumonia risk.	MedPath: GNN augmented with medical knowledge paths. Workflow: GRU EHR encoding → KG path retrieval → Path embedding via GNN → Attention-weighted path fusion → Binary risk classification.	Heart failure AUC: 0.934 (+3.2% over GRAM baseline). Pneumonia AUC: 0.911 (+2.8%).	Explicit medical knowledge paths from a clinical KG substantially improve EHR-based risk prediction beyond state-of-the-art attention baselines. Validates the KG-augmented clinical GNN premise.	MIMIC-III restricted to ICU patients — limited outpatient generalizability. KG path selection is heuristic; not all medically meaningful paths captured.	Automated KG path discovery using reinforcement learning. Extend to symptom-driven inference for primary care settings.
15	Sun, Li & Yang (2021) Expert Systems with Applications	Existing symptom-disease diagnostic models treat all disease-symptom relationships as unweighted binary connections, ignoring co-occurrence frequency and clinical severity.	Columbia University Disease-Symptom Knowledge Base (same source as this project): 41 diseases, 131 symptoms, 4,920 patient records.	Bipartite GCN for disease-symptom link prediction. Workflow: Binary bipartite graph construction → GCN message passing (unweighted) → Dot-product link prediction → Top-k disease ranking.	Top-1 accuracy: 78.3%. Top-3 accuracy: 91.2%. Binary GCN outperforms SVM (+12%) and Random Forest (+8%).	GCN-based link prediction outperforms tabular ML for symptom-disease diagnosis. Most directly related prior work — uses identical dataset but binary (unweighted) edges, the primary research gap this project closes.	Binary unweighted edges lose probability and severity information. No XAI module — model is a black box. Synthetic dataset only.	Integrate probability-and-severity edge weights for prediction granularity. Add XAI module for clinical justification of predictions.
16	Yuan, Yu, Gui & Ji (2023) IEEE TPAMI	Dozens of GNN explainability methods existed with no unified taxonomy, making it impossible to compare approaches or identify research gaps.	Survey of 60+ GNN explainability papers on molecular (MUTAG, NCI1), social (BA-Shapes), and biomedical (drug interaction) graph benchmarks.	Taxonomic literature survey. Workflow: Method classification (instance-level vs. model-level; gr adient/perturbation/decompositio n) → Benchmark evaluation → F aithfulness/Stability/Completenes s assessment.	Faithfulness: Gradient methods ~0.78, Perturbation ~0.71, Decomposition ~0.85 on molecular benchmarks.	Most GNN explainers fail to preserve domain-semantic meaning. Decomposition-based methods (including dot-product attribution) offer superior completeness. Directly motivates Latent Feature Attribution in this project.	Taxonomy based on 2023 literature; newer methods not covered. Benchmark datasets not representative of clinical complexity.	Clinical XAI benchmarks with physician-validated explanation quality metrics. Self-explainable GNN architectures producing intrinsic explanations.
17	Luo et al. (2020) NeurIPS 2020	GNNExplainer generates per-instance edge masks with high variance, instability, and inability to reveal global model patterns.	BA-Shapes (synthetic node classification), BA-Community, MUTAG (mutagenicity molecular graphs), Graph-SST2 (sentiment graphs).	PGExplainer: parameterized global edge mask generator conditioned on node embeddings. Workflow: GNN training → PGExplainer network training → Edge importance mask generation → Fidelity/sparsity/stability evaluation.	Explanation accuracy: BA-Shapes 92.3% vs. GNNExplainer 77.1%. Stability: PGExplainer 5× more stable than GNNExplainer.	Parameterized global explainers produce more faithful and stable explanations than per-instance approaches. Provides strong baseline comparison for our Latent Feature Attribution method.	Requires training a secondary explanation model (additional compute cost). Explanation quality degrades on very large, dense graphs.	Apply to biomedical link prediction with clinical validation of explanation fidelity. Combine parameterized explainers with semantic domain constraints.
18	Amann et al. (2020) BMC Medical Informatics and Decision Making	AI clinical systems generate predictions without justifications, preventing clinician trust, regulatory approval, and ethical accountability.	Multi-stakeholder qualitative study: 40+ expert interviews with clinicians, regulators, patients, AI developers, and ethicists in Europe and North America.	Qualitative research (grounded theory). Workflow: Expert interviews → Thematic coding → Stakeholder need taxonomy → Explainability requirement formalization.	Five distinct stakeholder groups with different explainability needs. Trust threshold: clinicians require causal, domain-aligned explanations for adoption.	Domain-aligned, mathematically traceable explanations are simultaneously a regulatory requirement and an ethical imperative. Formalizes stakeholder needs that our XAI module directly addresses.	Qualitative study — not statistically generalizable. Jurisdictional variation in AI regulation not fully captured.	Quantitative trials measuring XAI impact on diagnostic accuracy and clinician trust. Legal framework development for AI explanation requirements (EU AI Act, FDA SaMD).

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19	Jiménez-Luna, Grisoni & Schneider (2020) Nature Machine Intelligence	Drug discovery AI models produce accurate predictions but without clinical rationale, making it impossible for medicinal chemists to trust or act on model outputs.	Multiple drug discovery benchmarks: ESOL (solubility), Tox21 (toxicity), BBBP (blood-brain barrier), HIV activity datasets.	Survey of XAI methods for molecular property prediction. Workflow: Attribution method selection (GNNExplainer, SHAP, attention, gradient saliency) → Application to molecular GNNs → Practitioner interpretability evaluation.	Attribution fidelity (deletion AUC): SHAP 0.81, GNNExplainer 0.77, attention weights 0.69.	Attribution-based explanation methods provide actionable, practitioner-interpretable insights in drug discovery. Validates the conceptual basis of Latent Feature Attribution for clinical AI.	Attribution quality is task-dependent; no universal explainability metric exists. Human evaluation of explanation quality is expensive and not standardized.	Apply attribution-based XAI to clinical disease prediction models. Develop standardized benchmarks for clinical XAI quality assessment.
20	Tjoa & Guan (2021) IEEE TNNLS	Healthcare AI models are black boxes. Clinicians, patients, and regulators cannot understand, validate, or trust model decisions, blocking clinical deployment.	Survey of 250+ XAI papers across medical imaging (X-ray, MRI, histopathology), clinical risk prediction, EHR, and genomics applications.	Systematic literature survey and classification. Workflow: XAI method taxonomy (attention/gradient/perturbation/concept/example-based) → Medical domain mapping → Multi-criteria evaluation (fidelity, interpretability, stability, completeness).	Multi-criteria scores across XAI methods. Completeness identified as most critical criterion. Gradient (Fidelity: High), Attention (Interpretability: High, Completeness: Low).	Completeness is the most critical XAI criterion for clinical deployment. Our Latent Feature Attribution satisfies completeness exactly (Eq. 10). Establishes research agenda for transparent, trustworthy clinical AI.	Survey rapidly outdated. Clinical validation of XAI methods remains scarce.	Self-explainable clinical GNNs generating explanations without post-hoc processing. Regulatory standardization of XAI evaluation criteria for medical AI certification.

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